A Bidirected-Graph Theory of the Standard Nuclear oliGARCHy

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Abstract

We develop a graph-theoretic model of strategic interaction among nine districts, each occupying one of four states: Peace, Experiment, War, and Stagnation. Vertices of the resulting 36-vertex graph correspond to district-state pairs; edges, termed *technologies*, are partitioned into three families: local (intra-district) arcs, ordinary relational (unilateral, inter-district) arcs, and joint (bidirected, coupled inter-district) edges, the last requiring the strict Edmonds–Johnson formalism of bidirected graphs to represent correctly.

We furnish the model with algebraic, logical, game-theoretic, welfare-economic, physical, chemical, epidemiological, statistical, psychological, political-scientific, and computational readings, proving several structural propositions along the way. The result is a compact but genuinely interdisciplinary formal object: a single 36-vertex bidirected graph capacious enough to host analyses ordinarily distributed across a dozen separate literatures.

The paper ends with "The End"

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1 Introduction

Coupled political units routinely face a choice not merely of *whether* to escalate or de-escalate, but of *which instrument* – domestic policy, unilateral coercion, or binding bilateral agreement – to use in doing so. This paper formalizes that choice as a graph on 36 vertices, in which every available instrument is an edge and every edge is a named *technology*. Section 2 fixes notation. Section 3 introduces the three edge families and argues that the joint family cannot be represented as ordinary arcs, motivating the bidirected-graph formalism of Edmonds and Johnson [1,2]. Section 4 gives the algebraic representation. Section 5 recasts the graph as a Kripke frame [4]. Sections 6–11 develop, in turn, game-theoretic and welfare-economic, warfare-economic, statistical, physical/chemical, epidemiological, psychological, and political-scientific readings of the same object. Section 12 treats algorithmic and data-scientific questions the graph raises. Section 13 discusses scope and limitations.

2 The State Space

Let $D = \{d_1, \ldots, d_9\}$ be a set of nine *districts* and let

$$\Sigma = \{P, E, W, S\}$$

be a set of four *states*: Peace, Experiment, War, and Stagnation. The object of study is the vertex set

$$V = D \times \Sigma, \qquad |V| = 9 \times 4 = 36.$$

A vertex (d_i, σ) represents district d_i occupying state σ. We emphasize that V does *not* track the joint configuration of all nine districts simultaneously: that finer object would be Σ^9, of cardinality $4^9 = 262{,}144$. Collapsing to the 9×4 product is a deliberate modeling choice, revisited in Section 13, that trades the ability to condition a district's transition on every other district's exact state for a vertex set small enough to admit exact algebraic and logical analysis.

3 Technologies as Edges

We call the edges of our graph *technologies* and partition the technology set $\mathcal{T}$ into three disjoint families,

$$\mathcal{T} \;=\; \mathcal{T}_{\text{loc}} \;\sqcup\; \mathcal{T}_{\text{ord}} \;\sqcup\; \mathcal{T}_{\text{joint}},$$

according to whether a technology is local to one district, unilateral between two districts, or genuinely joint between two districts.

3.1 Local technologies

Definition 1. A *local technology* is an ordinary arc $(d_i, \sigma) \to (d_i, \sigma')$, $\sigma \neq \sigma'$, changing only district d_i's own state.

Assuming every district has access to the same tree of local technologies (an assumption we relax in Section 13), $\mathcal{T}_{\text{loc}}$ decomposes into nine disjoint copies of a single 4-vertex digraph G_0 on Σ. We take as illustrative default the *complete* digraph on Σ, i.e. every one of the twelve ordered pairs is realized by a named technology:

Table 1: An illustrative complete local technology tree G_0.

Transition	Technology
$P \to E$	Research Initiative
$P \to W$	Preemptive Strike
$P \to S$	Austerity
$E \to P$	Disarmament
$E \to W$	Weaponization
$E \to S$	Program Cancellation
$W \to P$	Armistice
$W \to E$	Wartime Research
$W \to S$	Attrition
$S \to P$	Reconstruction
$S \to E$	Innovation Grant
$S \to W$	Desperate Gambit

Since G_0 is the complete digraph on 4 vertices, $|\mathcal{T}_{\mathrm{loc}}| = 9 \times 12 = 108$.

3.2 Ordinary relational technologies

Definition 2. An *ordinary relational technology* is an arc $(d_i, \sigma) \to (d_j, \sigma')$, $i \neq j$, representing district d_i unilaterally triggering district d_j's transition into σ'.

We read such an arc as a trigger rather than a full transition record: it does not encode d_j's prior state, in keeping with the 36-vertex resolution's deliberate silence on joint configurations. Representative technologies: Proliferation, $(d_i, E) \to (d_j, E)$; Coercion, $(d_i, W) \to (d_j, W)$; Sanctions, $(d_i, W) \to (d_j, S)$. The space of possible ordinary relational arcs has cardinality $9 \times 8 \times 4 \times 4 = 1152$, an order of magnitude larger than the local layer.

3.3 Joint technologies and the bidirected formalism

Some technologies – treaties, joint ventures, coordinated stand-downs – have no unilateral actor: neither district is the "source" and the other the "target." Representing such a technology as two ordinary arcs would misstate it, since two ordinary arcs let either district invoke its half alone. The correct object is a *bidirected* edge, in the sense of Edmonds and Johnson [2, 3]: each endpoint of an edge is independently marked *tail* or *head*, giving three edge types instead of one:

1. an *ordinary arc*, one tail and one head;
2. a *both-tails* edge, both endpoints tails;
3. a *both-heads* edge, both endpoints heads.

Definition 3. A *joint technology* linking districts d_i, d_j is a linked pair of bidirected edges:

- a both-tails edge over the source pair $\{(d_i, \sigma), (d_j, \sigma)\}$, both endpoints simultaneously feeding into the technology;
- a both-heads edge over the destination pair $\{(d_i, \sigma'), (d_j, \sigma')\}$, both endpoints simultaneously arriving from it.

Example 1. *Mutual Disarmament Treaty*: both-tails on $\{(d_i, W), (d_j, W)\}$, both-heads on $\{(d_i, P), (d_j, P)\}$.

The payoff is not that a static graph enforces simultaneity by itself – no graph does – but that this shape is the correct one on which to later hang a rule of the form "fires only when every tail is satisfied, arrives at every head together." Figure 1 depicts all three edge types across two districts.

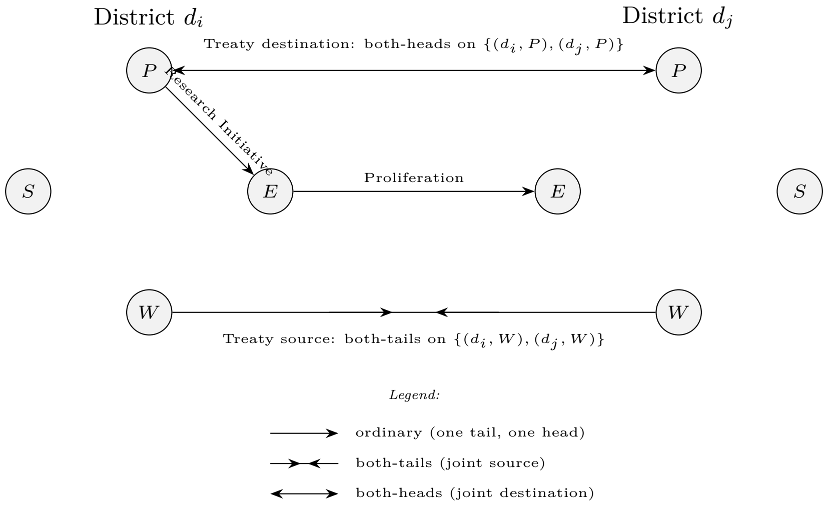

Figure 1: Local, ordinary relational, and joint technologies between two districts.

4 Matrix and Algebraic Representations

Definition 4. The *signed incidence matrix* of a bidirected graph on vertex set V with edge set $\mathcal{T}$ is $\mathbf{M} \in \{-1, 0, 1\}^{|V|\times|\mathcal{T}|}$, with $M_{v,\tau} = -1$ if v is a tail-endpoint of τ, $M_{v,\tau} = +1$ if v is a head-endpoint of τ, and $M_{v,\tau} = 0$ otherwise.

An ordinary arc's column has exactly one -1 and one $+1$; a both-tails column has two -1's; a both-heads column has two $+1$'s. This is, precisely, the standard formal definition of a bidirected graph.

Lemma 1. *Let $A_0 \in \{0, 1\}^{4\times4}$ be the adjacency matrix of G_0, let $B_{ij} \in \{0, 1\}^{4\times4}$ record the ordinary relational arcs from d_i to d_j, and let $U_{ij} \in \{0, 1\}^{9\times9}$ be the matrix unit with a 1 in position (i, j) and 0 elsewhere. Restricting to $\mathcal{T}_{\text{loc}} \cup \mathcal{T}_{\text{ord}}$, the resulting 36×36 ordinary 0/1 adjacency matrix satisfies*

$$\mathbf{A} \;=\; I_9 \otimes A_0 \;+\; \sum_{i\neq j} U_{ij} \otimes B_{ij}.$$

Proof. Index V by pairs (i, σ) so that $\mathbf{A}$ is a 9×9 array of 4×4 blocks. By definition of the Kronecker product, block (k, l) of $I_9 \otimes A_0$ equals $(I_9)_{kl}A_0$, which is A_0 if $k = l$ and the zero matrix otherwise; block (k, l) of $U_{ij} \otimes B_{ij}$ equals $(U_{ij})_{kl}B_{ij}$, which is B_{ij} if $(k, l) = (i, j)$ and zero otherwise. Summing the second term over all $i \neq j$ therefore places B_{ij} exactly in off-diagonal block (i, j) for every ordered pair, while the diagonal blocks receive only the first term, A_0. This is exactly the block structure of $\mathbf{A}$: diagonal blocks encode local arcs within a district, off-diagonal blocks encode ordinary relational arcs between districts. ■ □

Corollary 1. *Since G_0 is complete, $A_0 = J_4 - I_4$, where J_4 is the all-ones matrix; hence $I_9 \otimes A_0 = I_9 \otimes J_4 - I_{36}$.*

Joint technologies are exactly what falls outside Lemma 1: no single entry of an ordinary adjacency matrix can mean "these two vertices, together, are the source." The full picture requires the incidence matrix $\mathbf{M}$, not $\mathbf{A}$.

5 Logical Foundations: The Graph as a Kripke Frame

Definition 5. A *Kripke frame* [4] is a pair (W, R), W a set of possible worlds and $R \subseteq W \times W$ an accessibility relation. Given (W, R) and a valuation of propositional variables at worlds, $\Diamond\varphi$ holds at w iff φ holds at some w' with wRw'; $\Box\varphi$ holds at w iff φ holds at every such w'.

Our graph restricted to $\mathcal{T}_{\text{loc}} \cup \mathcal{T}_{\text{ord}}$ is precisely a Kripke frame on $W = V$, with R the arc relation. Joint technologies, however, do not fit: a both-tails or both-heads edge is not an ordered pair $(u, v) \in W \times W$ with a single accessed world, but a coupling of two worlds neither of which is uniquely "source." Capturing $\mathcal{T}_{\text{joint}}$ within modal logic proper would require a genuinely richer semantics (e.g. a team- or hyper-relational semantics), mirroring exactly the reason ordinary digraphs were insufficient in Section 3: solo action fits the standard formalism; joint action does not.

Proposition 1. *Under the complete illustrative G_0, for any district d_i and any $\sigma, \sigma' \in \Sigma$, $\Diamond$ applied once at (d_i, σ) reaches (d_i, σ') whenever $\sigma \neq \sigma'$.*

Proof. Immediate: completeness of G_0 means every ordered pair (σ, σ'), $\sigma \neq \sigma'$, is an arc of G_0, hence of the local copy at d_i. ■ □

Proposition 2. *Let G_0 be strongly connected and define the quotient digraph Q on D by $d_i \to d_j$ in Q iff some ordinary relational arc runs from a state of d_i to a state of d_j. If Q is strongly connected, the full graph on $(V, \mathcal{T}_{\text{loc}} \cup \mathcal{T}_{\text{ord}})$ is strongly connected.*

Proof. Any two vertices within the same district are mutually reachable, since each district's local copy of G_0 is strongly connected. For (d_i, σ) and (d_j, σ') in distinct districts, strong connectivity of Q gives a walk $d_i = d_{k_0} \to d_{k_1} \to \cdots \to d_{k_m} = d_j$; each step $d_{k_t} \to d_{k_{t+1}}$ corresponds to at least one relational arc from some specific state of d_{k_t} to some specific state of $d_{k_{t+1}}$. Because every district's local subgraph is strongly connected, one may always first move, using local arcs alone, from the current state at d_{k_t} to the specific state required as the source of the next relational arc, traverse that arc, and repeat. Concatenating these segments yields a walk from (d_i, σ) to (d_j, σ'); since i, j, σ, σ' were arbitrary, the graph is strongly connected. ■ □

6 Game-Theoretic and Welfare-Economic Interpretation

Assign each state a flow utility, e.g. $u(P) = 2$, $u(E) = 0$, $u(W) = -3$, $u(S) = -1$ (illustrative). For a joint configuration $x \in \Sigma^9$, define utilitarian welfare $U(x) = \sum_{i=1}^{9} u(x_i)$ or Rawlsian welfare $U_{\text{R}}(x) = \min_i u(x_i)$ [7]. A joint technology moving two districts from W to P is a strict Pareto improvement for both, *provided* both consent; the entire motivation for $\mathcal{T}_{\text{joint}}$ in Section 3 is that a graph built only from ordinary arcs cannot force that joint consent.

The tension this resolves is a security dilemma with the structure of a Prisoner's Dilemma [5, 6]. Let each district choose Escalate or De-escalate:

Table 2: Payoffs (district d_i, district d_j) under unilateral escalation choices.

	d_j: De-escalate	d_j: Escalate
d_i: De-escalate	$(3, 3)$	$(0, 5)$
d_i: Escalate	$(5, 0)$	$(1, 1)$

With $5 > 3 > 1 > 0$ (temptation > reward > punishment > sucker's payoff), Escalate strictly dominates De-escalate for each district individually, so the unique Nash equilibrium is mutual escalation $(1, 1)$, though $(3, 3)$ Pareto-dominates it. Ordinary arcs alone – e.g. two independent local Armistice arcs – cannot escape this trap, since either district could always defect back to Escalate; a joint technology requiring simultaneous both-tails commitment is precisely the instrument that removes the possibility of unilateral defection at the moment of signing, which is the graph-theoretic content of "a treaty."

7 Warfare Economics

Let $x(t), y(t)$ denote the war-fighting capacities of two districts locked in the W state, with aimed-fire ("modern") combat dynamics of Lanchester's square law [15]:

$$\frac{dx}{dt} = -\beta y, \qquad \frac{dy}{dt} = -\alpha x.$$

Multiplying the first equation by αx and the second by βy gives $\alpha x\, \dot{x} = \beta y\, \dot{y} = -\alpha\beta xy$, so

$$\frac{d}{dt}\Big(\alpha x(t)^2 - \beta y(t)^2\Big) = 0 \quad \Longrightarrow \quad \alpha x(t)^2 - \beta y(t)^2 = \alpha x_0^2 - \beta y_0^2$$

for all t: the celebrated *square-law invariant*. Capacity decays monotonically under W, motivating why the Attrition arc $W \to S$ is, empirically, so often taken: prolonged war is a self-terminating process even absent an external Armistice. The cumulative economic burden of remaining at W over $[0, T]$ is $C(T) = \int_0^T c(t)\, dt$ for instantaneous expenditure $c(t)$, the opportunity cost side of the classical guns-versus-butter tradeoff.

8 Statistical Dynamics: GARCH-Type Transition Volatility

Let $p_{\tau,t}$ denote the probability that technology $\tau \in \mathcal{T}$ fires at discrete time t, driven by a latent shock $\varepsilon_{\tau,t} = \sigma_{\tau,t} z_{\tau,t}$ with $z_{\tau,t}$ i.i.d., mean zero, unit variance, and conditional variance following Engle and Bollerslev's GARCH(1, 1) process [13, 14]:

$$\sigma^2_{\tau,t} = \omega_\tau + \alpha_\tau\, \varepsilon^2_{\tau,t-1} + \beta_\tau\, \sigma^2_{\tau,t-1}, \qquad \omega_\tau > 0,\ \alpha_\tau, \beta_\tau \geq 0,\ \alpha_\tau + \beta_\tau < 1,$$

the last inequality ensuring covariance stationarity, with unconditional variance $\omega_\tau/(1 - \alpha_\tau - \beta_\tau)$. Firing probability follows a logistic link, $p_{\tau,t} = \Lambda(\mu_\tau + \varepsilon_{\tau,t})$. Illustrative estimates:

Table 3: Illustrative GARCH(1, 1) parameters by technology class.

Technology class	ω	α	β
Coercion	0.02	0.10	0.85
Proliferation	0.03	0.06	0.80
Armistice	0.05	0.15	0.60
Mutual Disarmament Treaty	0.04	0.05	0.70

The point of this layer – and of the name *oliGARCHy* – is that crises cluster: a high realization of $\sigma^2_{\tau,t-1}$ for a Coercion-type edge raises the expected volatility of the *next* period's shock, producing exactly the volatility clustering long documented in financial return series, now reinterpreted as clustering of escalation risk. Parameters would, in an applied setting, be fit by maximum likelihood on panel data of realized district transitions.

9 Physical, Chemical, and Epidemiological Analogies

9.1 A Potts-model reading

Treat each district as a site carrying a spin $\sigma_i \in \Sigma$: a 4-state Potts model [17]. With coupling J over interacting (relationally connected) district pairs $\langle i, j\rangle$ and a field term $h(\sigma)$,

$$H(\sigma) = -J \sum_{\langle i,j\rangle} \delta_{\sigma_i,\sigma_j} - \sum_i h(\sigma_i), \qquad \Pr(\sigma) = \frac{e^{-H(\sigma)/k_B T}}{Z}, \qquad Z = \sum_{\sigma \in \Sigma^9} e^{-H(\sigma)/k_B T}.$$

As background "temperature" T – read here as the intensity of the volatility clustering of Section 8 – rises, the system moves from an ordered phase (districts sharing one state) toward a disordered phase (no consensus), a bona fide order-disorder phase transition in the sense of Potts's original paper.

9.2 An Arrhenius reading

Individual technologies may be assigned an activation-energy-style threshold: $k_\tau = A_\tau \exp(-E_a^{(\tau)}/k_B T)$, where $E_a^{(\tau)}$ stands for the diplomatic or economic barrier to invoking τ. Rising T lowers effective barriers uniformly, exactly as rising temperature accelerates a chemical reaction.

9.3 An epidemiological reading

Map P, E, W, S onto a Susceptible–Exposed–Infected–Recovered (SEIRS) compartmental system [16], with recovered districts eventually returning to Peace (waning immunity). Writing p, e, w, s for the district fractions in each state:

$$\frac{dp}{dt} = -\lambda p w + \gamma s, \quad \frac{de}{dt} = \lambda p w - \kappa e, \quad \frac{dw}{dt} = \kappa e - \mu w, \quad \frac{ds}{dt} = \mu w - \gamma s.$$

Summing the four right-hand sides gives zero, confirming $p + e + w + s \equiv 1$ is preserved. Figure 2 depicts the cycle. Linearizing about the war-free equilibrium $(p, e, w, s) = (1, 0, 0, 0)$ with transmission matrix $F = \begin{pmatrix} 0 & \lambda \\ 0 & 0 \end{pmatrix}$ and transition matrix $V = \begin{pmatrix} \kappa & 0 \\ -\kappa & \mu \end{pmatrix}$ for the (e, w) block, the next-generation matrix is

$$FV^{-1} = \begin{pmatrix} 0 & \lambda \\ 0 & 0 \end{pmatrix} \frac{1}{\kappa\mu} \begin{pmatrix} \mu & 0 \\ \kappa & \kappa \end{pmatrix} = \begin{pmatrix} \lambda/\mu & \lambda/\mu \\ 0 & 0 \end{pmatrix},$$

whose spectral radius gives the basic reproduction number $R_0 = \lambda/\mu$: war-contagion via Coercion-type arcs takes hold precisely when $R_0 > 1$.

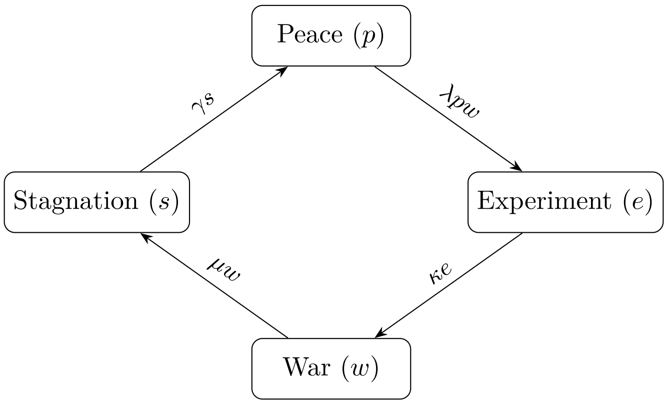

Figure 2: The SEIRS compartmental reading of the state space.

10 Psychological Dimensions

Decision-makers choosing between Escalate and De-escalate in Table 2 are unlikely to be the risk-neutral expected-utility maximizers of Section 6. Cumulative prospect theory [8] posits a value function, referenced to a status-quo point,

$$v(x) = \begin{cases} x^{\alpha} & x \geq 0 \\ -\lambda(-x)^{\beta} & x < 0 \end{cases},$$

with commonly estimated $\alpha, \beta \approx 0.88$ and loss-aversion coefficient $\lambda \approx 2.25$. Losses looming larger than equivalent gains makes de-escalating – perceived as forgoing a possible future gain from unilateral escalation, or worse, as accepting a loss relative to a war-footing reference point – feel disproportionately costly, reinforcing the stickiness of the mutual-escalation equilibrium found in Section 6. At the collective level, the persistence of S once entered is consistent with the classical psychological finding of learned helplessness: repeated unsuccessful attempts to exit a state can raise the effective threshold required to try again, a hysteresis effect that here manifests as S typically requiring an explicit Reconstruction or Innovation Grant technology rather than passive drift. We use these terms as heuristic, population-level analogies for calibrating transition thresholds, not as clinical claims about any individual or institution.

11 Political Science, Geopolitics, and International Relations

The three technology families of Section 3 map naturally onto the major paradigms of international-relations theory.

Table 4: IR paradigms and the technology families they emphasize.

Paradigm	Emphasized family	Mechanism
Realism [10]	$\mathcal{T}_{\text{ord}}$	anarchy, security dilemma, Coercion
Liberal institutionalism [11]	$\mathcal{T}_{\text{joint}}$	treaties escape the Prisoner's Dilemma trap
Constructivism [12]	graph structure itself	norms determine which arcs exist at all

Realist analysis takes $\mathcal{T}_{\text{ord}}$ as given and asks which equilibrium rational, self-interested districts reach (Section 6). Liberal institutionalism explains the *value added* by $\mathcal{T}_{\text{joint}}$: binding institutions are valuable precisely because they alter what is achievable, not merely what is chosen. Constructivism operates one level up, on the graph itself: a district that has internalized a norm against unprovoked first strikes simply does not have the arc $P \to W$ available in its local copy of G_0, a change to $\mathcal{T}_{\text{loc}}$ rather than to any payoff. A minimal just-war restriction can be encoded similarly, as a permissibility subgraph $\mathcal{T}^{\text{jwt}} \subseteq \mathcal{T}_{\text{ord}} \cup \mathcal{T}_{\text{joint}}$ admitting only arcs into W that satisfy last-resort and proportionality side-conditions. Finally, Schelling-style deterrence [9] operates on the GARCH baseline of Section 8: a district's credible retaliatory capacity from W lowers μ_τ for Coercion-type arcs directed at it, a direct link between a classical strategic-stability argument and a modern volatility model.

12 Computational Methods: Algorithms, Complexity, and Data Science

Reachability. The set of vertices reachable from a given start vertex under $\mathcal{T}_{\mathrm{loc}} \cup \mathcal{T}_{\mathrm{ord}}$ is computed by breadth-first search in $O(|V| + |\mathcal{T}|)$ time:

```
BFS-REACHABLE(V, T, start):
visited <- {start}
queue <- [start]
while queue is not empty:
    u <- DEQUEUE(queue)
    for each edge (u -> v) in T:
        if v not in visited:
            visited <- visited + {v}
            ENQUEUE(queue, v)
return visited
```

Stochastic simulation. Sampling a trajectory of the continuous-time process whose edge rates are the GARCH-modulated intensities of Section 8 is an instance of the Gillespie stochastic simulation algorithm [18], originally developed for chemical reaction kinetics and applicable verbatim once technologies are read as reactions:

```
GILLESPIE-SSA(V, T, rates, t_max):
t <- 0; state <- initial configuration
while t < t_max:
    R <- sum of rates(tau) over all tau enabled at state
    if R = 0: break
    r1, r2 <- two independent Uniform(0,1) draws
    dt <- (1 / R) * ln(1 / r1)
    tau <- the enabled technology whose cumulative
           rate-share first exceeds r2 * R
    t <- t + dt
    state <- APPLY(state, tau)
return trajectory
```

Complexity of joint-technology selection. Suppose we wish to select the largest set of joint technologies (each coupling exactly two districts) that can fire simultaneously without any district being committed twice. This is exactly a maximum matching problem on the graph whose vertices are districts and whose edges are the available joint technologies, solvable in polynomial time by Edmonds's blossom algorithm [1]. This closes a loop opened in Section 3: adopting the Edmonds–Johnson bidirected formalism was not merely a representational convenience, since the associated optimization problem inherits a polynomial-time algorithm from the same body of theory. Richer variants – e.g. respecting a per-district budget constraint on how many treaties it may sign – typically lose this guarantee, inheriting NP-hardness from constrained-matching problems more generally.

Learning transition structure. A graph neural network over $(V, \mathcal{T})$ [19] updates a vector representation h_v at each vertex by aggregating over neighbors,

$$h_v^{(k+1)} = \phi\left(h_v^{(k)}, \bigoplus_{u\,:\,(u,v)\in\mathcal{T}} \psi\Big(h_u^{(k)}, h_v^{(k)}\Big)\right),$$

with ϕ, ψ learned functions and $\bigoplus$ a permutation-invariant aggregator, providing a data-driven route to predicting which technology a district is likeliest to invoke next, complementary to the parametric GARCH model of Section 8.

13 Discussion and Limitations

Two structural themes recur throughout. First, the 36-vertex resolution – $D \times \Sigma$ rather than Σ^9 – buys algebraic and logical tractability (Sections 4–5) at the cost of not tracking full joint configurations except where explicitly reintroduced (Sections 6 and 9). Second, ordinary structure – ordinary arcs, ordinary adjacency matrices, ordinary Kripke frames – suffices exactly for solo action; genuinely joint action requires the strictly richer bidirected/incidence-matrix formalism at every level examined, from the graph itself (Section 3) to its logic (Section 5) to its associated optimization problem (Section 12). All numerical parameters given – utilities, GARCH coefficients, Lanchester coefficients, Potts couplings – are illustrative placeholders, not empirical estimates. Finally, we note explicitly that this is a stylized, pedagogical formalism; it is not, and is not intended as, a tool for operational nuclear or military decision-making, domains whose real complexity and human stakes exceed anything captured here.

14 Conclusion

A 36-vertex graph, three families of technology-edges, and a strict reading of "bidirected" turn out to be enough scaffolding to host algebra, logic, game theory, welfare economics, warfare economics, statistical time series, statistical mechanics, chemical kinetics, epidemiology, decision psychology, and international-relations theory as so many interpretive layers on a single shared object. The unifying observation, restated once more, is simple: whatever the field, solo action is the easy case and joint action is the one that forces the richer formalism.

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Glossary

District (D) The set $\{d_1, \ldots, d_9\}$ of political units under study.

State space (Σ) The four-element set $\{P, E, W, S\}$: Peace, Experiment, War, Stagnation.

Vertex set (V) $V = D \times \Sigma$, the 36 district-state pairs.

Local technology ($\mathcal{T}_{\text{loc}}$) An ordinary arc changing one district's own state.

Ordinary relational technology ($\mathcal{T}_{\text{ord}}$) An arc by which one district unilaterally triggers another's transition.

Joint technology ($\mathcal{T}_{\text{joint}}$) A linked both-tails/both-heads bidirected edge pair representing action with no unilateral actor.

Bidirected graph A graph in which each edge-endpoint is independently marked tail or head, admitting ordinary arcs, both-tails, and both-heads edges.

Incidence matrix The $\{-1, 0, 1\}$-valued matrix recording, per edge, which vertices are tails (-1) and which are heads $(+1)$.

Kronecker product ($\otimes$) The block-matrix operation with $(A \otimes B)$ having block (i, j) equal to $A_{ij}B$.

Kripke frame A pair (W, R) of possible worlds and an accessibility relation, the standard semantics for modal logic.

GARCH Generalized Autoregressive Conditional Heteroskedasticity: a time-series model in which the conditional variance depends on past squared shocks and past variances, producing volatility clustering.

Lanchester's laws Differential-equation models of attrition in combat; the square law applies to aimed ("modern") fire.

Basic reproduction number (R_0) The expected number of secondary transitions caused by one case near a disease-free (here, war-free) equilibrium.

Potts model A statistical-mechanics spin model with $q \geq 2$ discrete states per site and nearest-neighbor coupling.

Nash equilibrium A strategy profile in which no player can improve its payoff by unilateral deviation.

Pareto efficiency An outcome such that no other outcome makes someone better off without making someone else worse off.

Prospect theory A descriptive theory of choice under risk featuring reference dependence and loss aversion.

Blossom algorithm Edmonds's polynomial-time algorithm for maximum matching in general (non-bipartite) graphs.

Security dilemma The realist mechanism by which one district's defensive measures are read as threatening by another, driving mutual escalation.

Just war theory The philosophical tradition specifying conditions (e.g. last resort, proportionality) under which resort to war is permissible.

The End